1 Planning (15 marks)

Defining the problem (3 marks)

P λ is the independent variable or vary λ. [1]

P θ is the dependent variable or measure θ (for each λ). [1]

P Light sources to be of similar intensity/brightness. [1]

Methods of data collection (5 marks)

M1 Labelled diagram showing observer, light sources with method of producing monochromatic light e.g. filter/coloured LED. [1]

M2 Method to measure wavelength: record from filter/LED or Young's slit/diffraction grating method. [1]

M3 Use a rule to measure the distances. [1]

M4 Method to determine θ, e.g. θ (or sin θ or tan θ) = separation/distance or

$$\tan\left(\theta/2\right) = \frac{\text{separation}}{2 \times \text{distance}}$$

Do not allow protractor methods. [1]

M5 Carry out the experiment in a dark room. [1]

Method of analysis (2 marks)

A Plot a graph of θ against λ. [Allow lg θ against lg λ]. [1]

A Relationship valid if straight line <u>through origin</u>. [1]
[If lg-lg then straight line with gradient = (+)1 (ignore reference to *y*-intercept)]

Safety considerations (1 mark)

S Lamp becomes hot, therefore do not touch/switch off when not in use or use gloves when moving hot lamp.

OR Light may damage eyes, therefore wear dark glasses or do not look at unprotected lamps.

[1]

Additional detail (4 marks)

D1/2/3/4 Relevant points might include [4]

1 Use vertical filament lamps. Allow vertical slits.
2 Additional detail on measuring λ e.g. use of equation for Young's slit/diffraction grating method.
3 Use of vernier calipers to measure the separation of light sources.
4 Use large distances/separations.
5 θ = sin θ = tan θ for small angles.
6 View with the same eye.
7 Method to ensure distances are perpendicular or observer <u>equidistant</u> from pair of lamps.
8 Repeat experiment for each λ <u>and</u> average.

Do not allow vague computer methods.

[Total: 15]

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